

NORMALIZATION EXAMPLE

Not in 1NF

emp_id	emp_name	emp_address	emp_mobile
101	Herschel	New Delhi	8912312390
102	Jon	Kanpur	8812121212 9900012222
103	Ron	Chennai	7778881212
104	Lester	Bangalore	9990000123 8123450987

1NF

emp_id	emp_name	emp_address	emp_mobile
101	Herschel	New Delhi	8912312390
102	Jon	Kanpur	8812121212
102	Jon	Kanpur	9900012222
103	Ron	Chennai	7778881212
104	Lester	Bangalore	9990000123
104	Lester	Bangalore	8123450987

Candidate Keys: {teacher_id, subject}
Non prime attribute: teacher_age

teacher_id	subject	teacher_age
111	Maths	38
111	Physics	38
222	Biology	38
333	Physics	40
333	Chemistry	40

2NF-Full functional dependency

teacher_id	teacher_age
111	38
222	38
333	40

teacher_details table:

teacher_id	subject
111	Maths
111	Physics
222	Biology
333	Physics
333	Chemistry

teacher_subject table:

Third Normal form (3NF)-Transitive dependency

emp_id	emp_name	emp_zip	emp_state	emp_city	emp_district
1001	John	282005	UP	Agra	Dayal Bagh
1002	Ajeet	222008	TN	Chennai	M-City
1006	Lora	282007	TN	Chennai	Urrapakkam
1101	Lilly	292008	UK	Pauri	Bhagwan
1201	Steve	222999	MP	Gwalior	Ratan

Candidate Keys: {emp_id}

Non-prime attributes: all attributes except emp_id are non-prime as they are not part of any candidate keys.

Here, emp_state, emp_city & emp_district dependent on emp_zip.

And, emp_zip is dependent on emp_id that makes non-prime attributes (emp_state, emp_city & emp_district) transitively dependent on super key (emp_id). This violates the rule of 3NF.

emp_id	emp_name	emp_zip
1001	John	282005
1002	Ajeet	222008
1006	Lora	282007
1101	Lilly	292008
1201	Steve	222999

employee table:

emp_zip	emp_state	emp_city	emp_district
282005	UP	Agra	Dayal Bagh
222008	TN	Chennai	M-City
282007	TN	Chennai	Urrapakka m
292008	UK	Pauri	Bhagwan
222999	MP	Gwalior	Ratan

Emp_zip:

Boyce Codd normal form (BCNF)

It is an advance version of 3NF

emp_id	emp_nationality	emp_dept	dept_type	dept_no_of_emp
1001	Austrian	Production and planning	D001	200
1001	Austrian	stores	D001	250
1002	American	design and technical support	D134	100
1002	American	Purchasing department	D134	600

Functional dependencies in the table above:

emp_id → emp_nationality
emp_dept → {dept_type, dept_no_of_emp}

Candidate key: {emp_id, emp_dept}

The table is not in BCNF as neither emp_id nor emp_dept alone are keys.

emp_id	emp_nationality
1001	Austrian
1002	American

emp_nationality table:

emp_id	emp_dept
1001	Production and planning
1001	stores
1002	design and technical support
1002	Purchasing department

emp_dept_mapping table:

emp_dept	dept_type	dept_no_of_emp
Production and planning	D001	200
stores	D001	250
design and technical support	D134	100
Purchasing department	D134	600

emp_dept table:

- **Functional dependencies:**

emp_id -> emp_nationality

emp_dept -> {dept_type, dept_no_of_emp}

- **Candidate keys:**

For first table: emp_id

For second table: emp_dept

For third table: {emp_id, emp_dept}

Sales Records

Cust ID	Cust Name	Item	Shipping Address	Newsletter	Supplier	Supplier Phone	Price
at_smith	Alan Smith	Xbox One	35 Palm St.	Xbox News	Microsoft	(800) Buy-XBOX	250
roger25	Roger Banks	PS4	47 Camp Rd.	PS News	Sony	(800) Buy-SONY	300
wilson44	Evan Wilson	Xbox One PS Vista	28 Rock Av.	Xbox News, PS News	Microsoft, Sony	(800) Buy-XBOX, (800) Buy-SONY	550
am_smith	Alan Smith	PS4	47 Camp Rd.	PS News	Sony	(800) Buy-SONY	300

1NF

Cust ID	Cust Name	Item	Shipping Address	Newsletter	Supplier	Supplier Phone	Price
at_smith	Alan Smith	Xbox One	35 Palm St.	Xbox News	Microsoft	(800) Buy-XBOX	250
roger25	Roger Banks	PS4	47 Camp Rd.	PS News	Sony	(800) Buy-SONY	300
wilson44	Evan Wilson	Xbox One	28 Rock Av.	Xbox News	Microsoft	(800) Buy-XBOX	250
wilson44	Evan Wilson	PS Vista	28 Rock Av.	PS News	Sony	(800) Buy-SONY	300
am_smith	Alan Smith	PS4	47 Camp Rd.	PS News	Sony	(800) Buy-SONY	300

2NF

Cust ID	Cust Name	Shipping Address	Newsletter	Item
at_smith	Alan Smith	35 Palm St.	Xbox News	Xbox One
roger25	Roger Banks	47 Camp Rd.	PS News	PS4
wilson44	Evan Wilson	28 Rock Av.	Xbox News	Xbox One
wilson44	Evan Wilson	28 Rock Av.	PS News	PS Vista
am_smith	Alan Smith	47 Camp Rd.	PS News	PS4

Item	Supplier	Supplier Phone	Price
Xbox One	Microsoft	(800) Buy-XBOX	250
PS4	Sony	(800) Buy-SONY	300
PS Vista	Sony	(800) Buy-SONY	300

3NF

Cust ID	Cust Name	Shipping Address	Newsletter	Item
at_smith	Alan Smith	35 Palm St.	Xbox News	Xbox One
roger25	Roger Banks	47 Camp Rd.	PS News	PS4
wilson44	Evan Wilson	28 Rock Av.	Xbox News	Xbox One
wilson44	Evan Wilson	28 Rock Av.	PS News	PS Vista
am_smith	Alan Smith	47 Camp Rd.	PS News	PS4

Item	Supplier	Price
Xbox One	Microsoft	250
PS4	Sony	300
PS Vista	Sony	300

Supplier	Supplier Phone
Microsoft	(800) Buy-XBOX
Sony	(800) Buy-SONY